Vacuum generation and monitoring

What is vacuum ?

A vacuum is defined as a diluted gas, or the corresponding state at which its pressure or density is lower than that of the ambient surrounding atmosphere. (Because atmospheric pressure fluctuates locally over the Earth's surface and lessens as altitude above sea level increases, it is not possible to specify a general upper limit for the vacuum range).

At the cruising altitude of a passenger jet, i.e. at approximately 10 km above the surface of the Earth, atmospheric pressure has already decreased to 290 mbar.

Weather balloon data are measured at an altitude of approximately 30 km at a pressure of 24 mbar.

An even greater distance from the Earth's surface, finally, brings us to satellite orbits at an altitude of 250 km, where the pressure has now decreased to 10^{-5} ï 10^{-6}. What prevails farther out in space is an ultra high vacuum of less than 10^{-14}.

In order to achieve a vacuum it is necessary to generate a pressure in a vessel that is lower than the ambient pressure.
Due to the Earth's gravity, atmospheric pressure varies with altitude in accordance with the barometric altitude formula:

Barometer formula

$$p = p_0 \cdot \exp\left(-\frac{g \cdot \rho_0 \cdot h}{p_0}\right)$$

Where:
Atmospheric pressure at sea level p0 = 1,013 mbar
Earth acceleration g = 9.81 m/s2
Density of air at sea level at 0 °C ρ0 = 1.293 kg/m3.

Combining the constants yields:

$$p = p_0 \cdot \exp\left(-\frac{h}{8{,}005\ m}\right)$$

If p = p0 / 2 and the formula is solved for h, the result is the half altitude value h½ = 5,548 m. In other words: Atmospheric pressure declines by one half every 5.548 km.

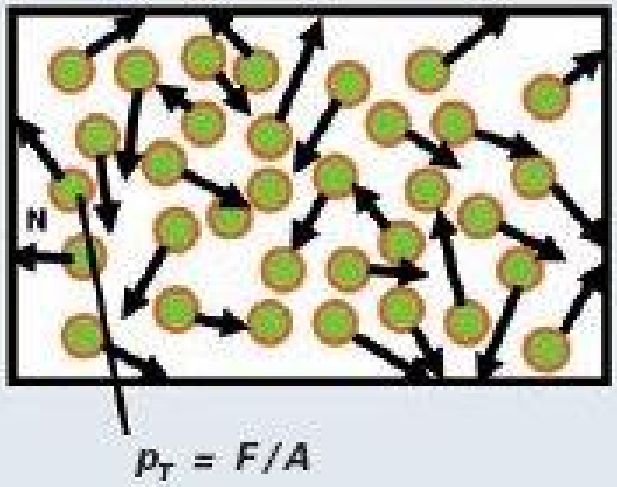
~ 10^{-18} hPa at 3K
≙ ~ $3 \cdot 10^{-18}$ g/m³
≙ ~ 100 particals/m³
Interstellar
BACKING VACUUM
Low Vacuum
Medium Vacuum
High Vacuum
Ultra High Vacuum
Pressure (hPa)
10^{-18}
10^{-17}
10^{-16}
10^{-15}
10^{-14}
10^{-13}
10^{-12}
10^{-11}
10^{-10}
10^{-9}
10^{-8}
10^{-7}
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
1
10
100
1000
Molecules in 1 cm³
10^{-2}
10^{-1}
1
10
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}
10^{7}
10^{8}
10^{9}
10^{10}
10^{11}
10^{12}
10^{13}
10^{14}
10^{15}
10^{16}
10^{17}
10^{18}
10^{19}
Mean Free Path
Path traveled by a molecule before striking another ither
10^{11}
10^{10}
10^{9}
10^{8}
10^{7}
10^{6}
10^{5}
10000
1000
100
10
1
100
10
1
10^{-1}
10^{-2}
10^{-3}
10^{-4}
Kilometer
Meter
Molecular Flow
Laminar Flow
Rotary vane pumps
Liquid ring pumps
Diaphragm pumps
Scroll pumps
Roots pumps
Screw pumps
Diffusion pumps
Turbomolecular pumps
Sputter ion getter pumps
Cryo pumps
Mechanical + capacitive measurement equipment
Thermal conductivity vacuum gauges
Penning vacuum gauges
Ionization vacuum gauges
Freeze drying – Packaging industry
Degasing, casting, dry vacuum smelting (super-pure metals)
Incandescent lamp manufacturing
Electronic tubes
Thin layers
Space simulation – Cryogenic research
Electron microscopy – Nuclear physics – Plasma physics – High energy physics
Particle accelerators – Storage rings
Cern ion trap
Between Earth and Moon
NGC891
Orion A. Nebula(H_2)
Visible interstellar gas nebulae
Space travel
ISS
Nothern Lights
– 1000 km
– 500 km
– 200 km
– 100 km
– 50 km
– 20 km
– 10 km
Weather
Air travel
10^{-15}
10^{-13}
10^{-11}
10^{-9}
10^{-7}
10^{-5}
10^{-3}
10^{-1}
10
1000
Pressure (hPa)
Operating Ranges of Major Vacuum Pumps
Vacuum Gauges
Typical Vacuum Applications
Space Travel Astro-Atmospheric Physics
Source: PSI
KU88-4/KI94/21.08.01

Pressure (hPa) Molecules in 1 cm³ Mean Free Path

Operating Ranges of Major Vacuum Pumps Vacuum Gauges

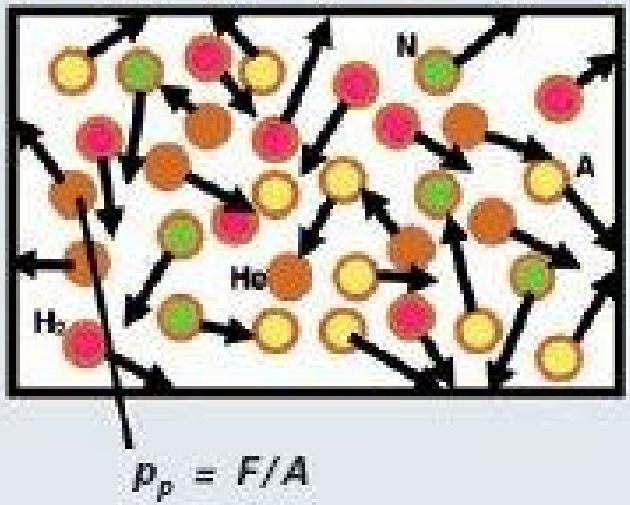
Typical Vacuum Applications

Space Travel Astro-Atmospheric Physics Pressure (hPa)



- Pressure in vessel
- One gas, e.g. nitrogen N_2
- Various gases, e.g. air

Total Pressure



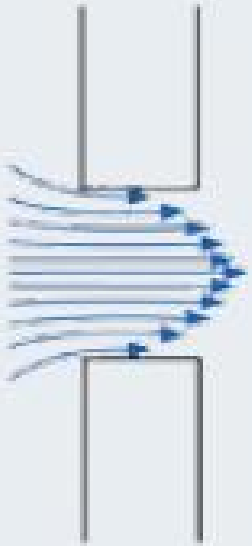
- Partial pressure
- Sum of partial pressures of all gases present equals total pressure
- All gases present evidenced through mass spectrometry

Partial Pressure

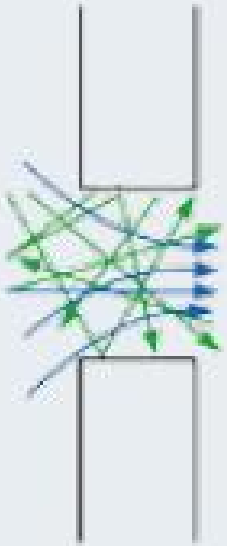
Pressure Range	Pressure / mbar	Molecular Number Density / cm^{-3}
Low vacuum	$10^{3} - 10^{0}$	$2.65 \cdot 10^{19} - 2.65 \cdot 10^{16}$
Medium vacuum	$10^{0} - 10^{-3}$	$2.65 \cdot 10^{16} - 2.65 \cdot 10^{13}$
High vacuum	$10^{-3} - 10^{-7}$	$2.65 \cdot 10^{13} - 2.65 \cdot 10^{9}$
Ultra high vacuum	$10^{-7} - 10^{-12}$	$2.65 \cdot 10^{9} - 2.65 \cdot 10^{4}$

Conversion Table for Pressure Units

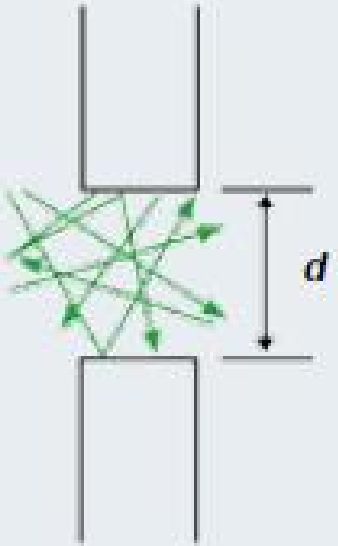
	Pa = N / m²	bar	mbar	µbar = dyn / cm²	Torr = mm Hg	micron µ = m Torr	atm	at	mm WS	psi = lbf / inch²	psf = lbf / ft²
Pa	1	$1 \cdot 10^{-5}$	$1 \cdot 10^{-2}$	10	$7.5 \cdot 10^{-3}$	7.5	$9.87 \cdot 10^{-6}$	$1.02 \cdot 10^{-5}$	0.102	$1.45 \cdot 10^{-4}$	$2.09 \cdot 10^{-2}$
bar	$1 \cdot 10^{5}$	1	$1 \cdot 10^{3}$	$1 \cdot 10^{6}$	750	$7.5 \cdot 10^{5}$	0.987	1.02	$1.02 \cdot 10^{4}$	14.5	$2.09 \cdot 10^{3}$
mbar	100	$1 \cdot 10^{-3}$	1	1,000	0.75	750	$9.87 \cdot 10^{-4}$	$1.02 \cdot 10^{-3}$	10.2	$1.45 \cdot 10^{-2}$	2.09
µbar	0.1	$1 \cdot 10^{-6}$	$1 \cdot 10^{-3}$	1	$7.5 \cdot 10^{-4}$	0.75	$9.87 \cdot 10^{-7}$	$1.02 \cdot 10^{-6}$	$1.02 \cdot 10^{-2}$	$1.45 \cdot 10^{-5}$	$2.09 \cdot 10^{-3}$
Torr	$1.33 \cdot 10^{2}$	$1.33 \cdot 10^{-3}$	1.33	1,330	1	1,000	$1.32 \cdot 10^{-3}$	$1.36 \cdot 10^{-3}$	13.6	$1.93 \cdot 10^{-2}$	2.78
micron	0.133	$1.33 \cdot 10^{-6}$	$1.33 \cdot 10^{-3}$	1.33	$1 \cdot 10^{-3}$	1	$1.32 \cdot 10^{-6}$	$1.36 \cdot 10^{-6}$	$1.36 \cdot 10^{-2}$	$1.93 \cdot 10^{-5}$	$2.78 \cdot 10^{-3}$
atm	$1.01 \cdot 10^{5}$	1.013	1,013	$1.01 \cdot 10^{6}$	760	$7.6 \cdot 10^{5}$	1	1.03	$1.03 \cdot 10^{4}$	14.7	$2.12 \cdot 10^{3}$
at	$9.81 \cdot 10^{4}$	0.981	981	$9.81 \cdot 10^{5}$	735.6	$7.36 \cdot 10^{5}$	0.968	1	$1 \cdot 10^{-4}$	14.2	$2.04 \cdot 10^{3}$
mm WC	9.81	$9.81 \cdot 10^{-5}$	$9.81 \cdot 10^{-2}$	98.1	$7.36 \cdot 10^{-2}$	73.6	$9.68 \cdot 10^{-5}$	$1 \cdot 10^{-4}$	1	$1.42 \cdot 10^{-3}$	0.204
psi	$6.89 \cdot 10^{3}$	$6.89 \cdot 10^{-2}$	68.9	$6.89 \cdot 10^{4}$	51.71	$5.17 \cdot 10^{4}$	$6.8 \cdot 10^{-2}$	$7.02 \cdot 10^{-2}$	702	1	144
psf	47.8	$4.78 \cdot 10^{-4}$	0.478	478	0.359	359	$4.72 \cdot 10^{-4}$	$4.87 \cdot 10^{-4}$	4.87	$6.94 \cdot 10^{-3}$	1



Continuous flow
$Kn < 0.01$
Low vacuum



Knudsen flow
$0.01 < Kn < 0.5$
Medium vacuum



Molecular flow
$Kn > 0.5$
High / Ultra high vacuum

- **Continuous flow**
  - Laminar, $Re < 2{,}300$, for circular pipes
  - Turbulent, $Re > 4{,}000$, for circular pipes
- **Knudsen flow**
- **Molecular flow**
- **Knudsen number** $Kn = \frac{\bar{l}}{d}$

$\bar{l}$ = Mean free path
l_{char} = Characteristic clearance of a component through which gas flows

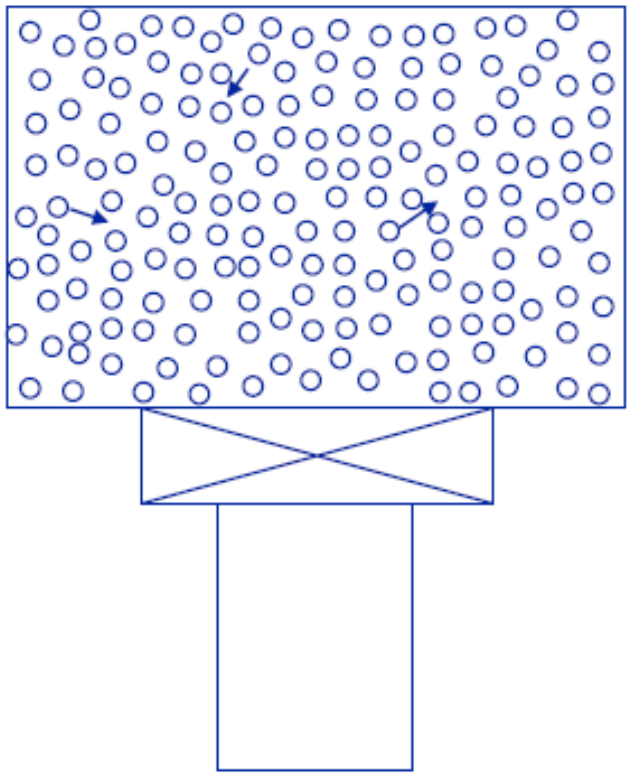
The dimensionless Knudsen number *Kn* is defined as the ratio between mean free path and component diameter

$$Kn = \frac{\bar{l}}{d}$$

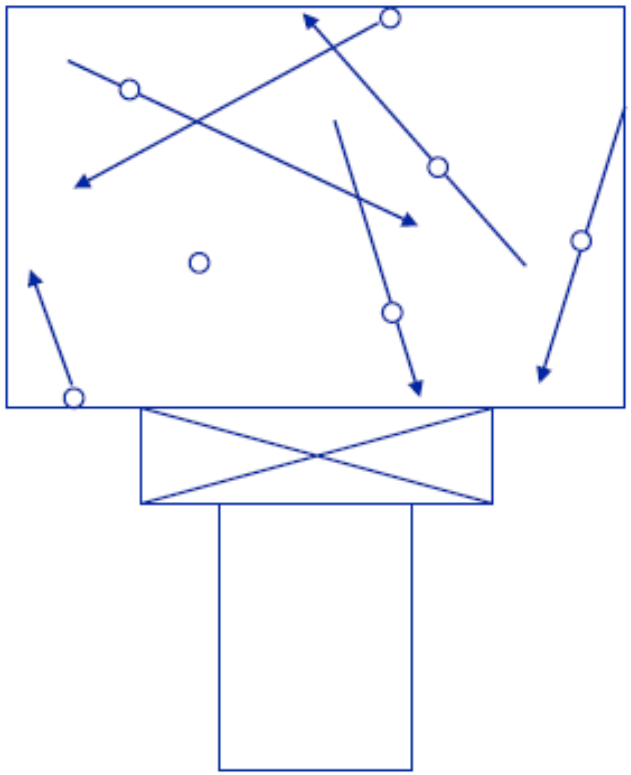
In the case of viscous flow, a distinction is made between laminar and turbulent flow. Laminar flow prevails at low flow speeds. At higher flow speeds, this changes to a turbulent flow. The occurrence of turbulent flow is contingent upon the Reynolds number

$$Re = \frac{\rho}{\eta} \cdot v \cdot d$$

Viscous and Molecular Flow



Viscous Flow
(momentum transfer between molecules)



Molecular Flow
(molecules move independently)

Continuous flow in low vacuum, p = 10^3 – 10^0 mbar, where *l* << *d*

What characterizes continuous flow, as well as viscous flow, is frequent contact between gas molecules, yet less frequent contact with the walls of the vessel. In this case, the mean free path of the gas molecules is significantly shorter than the dimensions *d* of the vacuum equipment.

Knudsen flow in medium vacuum, *p* = 10^0 – 10^{-3} mbar, where *l* ≤ *d*

If the Knudsen number is between 0.01 and 0.5, this is termed Knudsen flow. Because many process pressures are in the medium vacuum range, this type of flow occurs with corresponding frequency. Since this is a transitional flow, this range is transited relatively quickly when pumping down vacuum chambers. This means that the influence of this conductivity on pump-down times is correspondingly low.

Molecular flow in high vacuum, (p = 10^{-3} – 10^{-7} mbar), where *l* > *d* and in ultra high vacuum (*p* < 10^{-7} mbar), where *l* >> *d*

At Knudsen numbers of $Kn > 0.5$ molecular interaction virtually no longer occurs. What prevails is molecular flow. In this case, the product of pressure p and component diameter d is $p \cdot d \leq 1.3 \cdot 10^{-2}$ mbar · cm.

Classification of vacuum pumps

In connection with vacuum pumps, a distinction is made between gas-displacement vacuum pumps and gas-binding vacuum pumps.

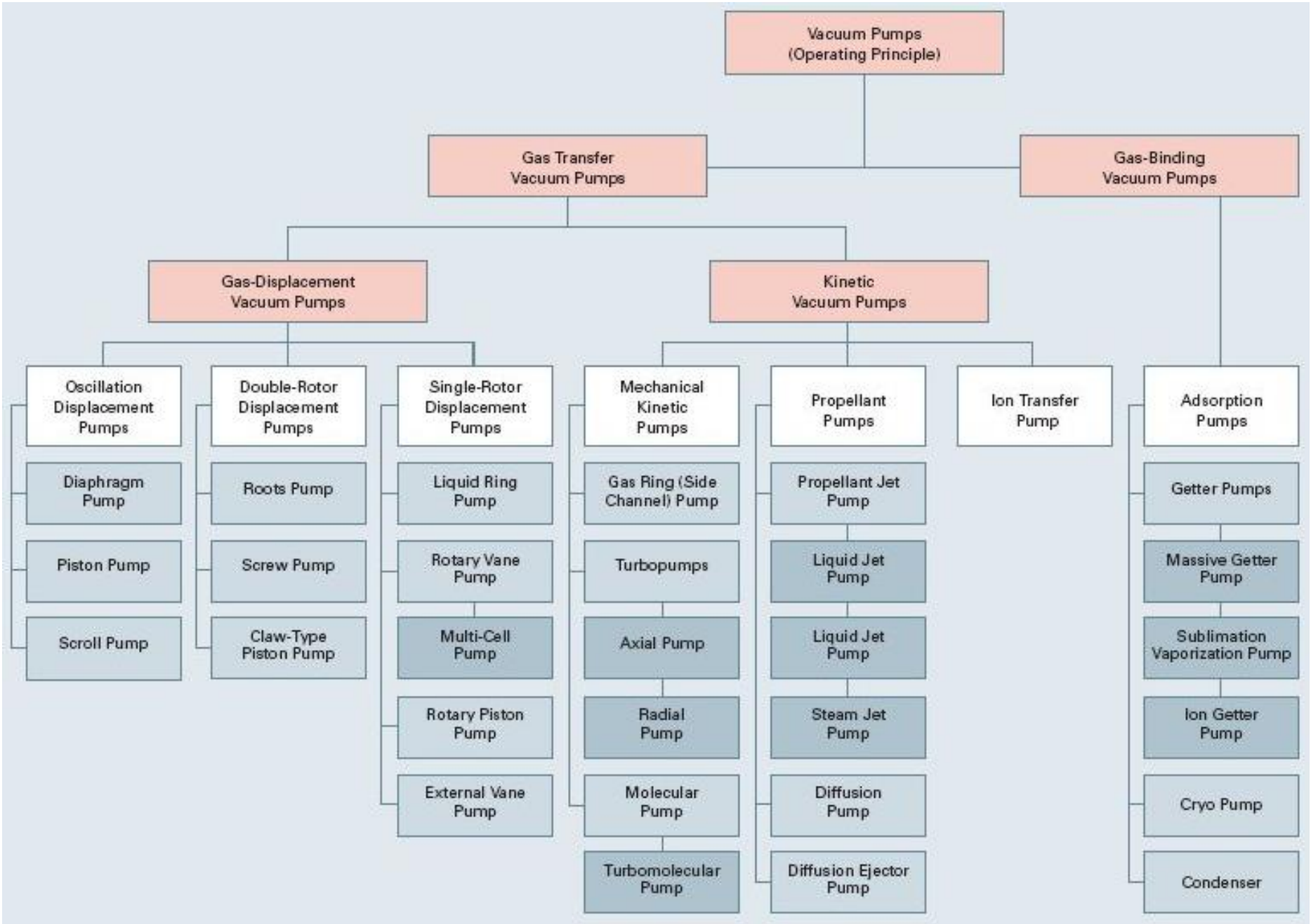
Gas-displacement vacuum pumps can be used without limitation

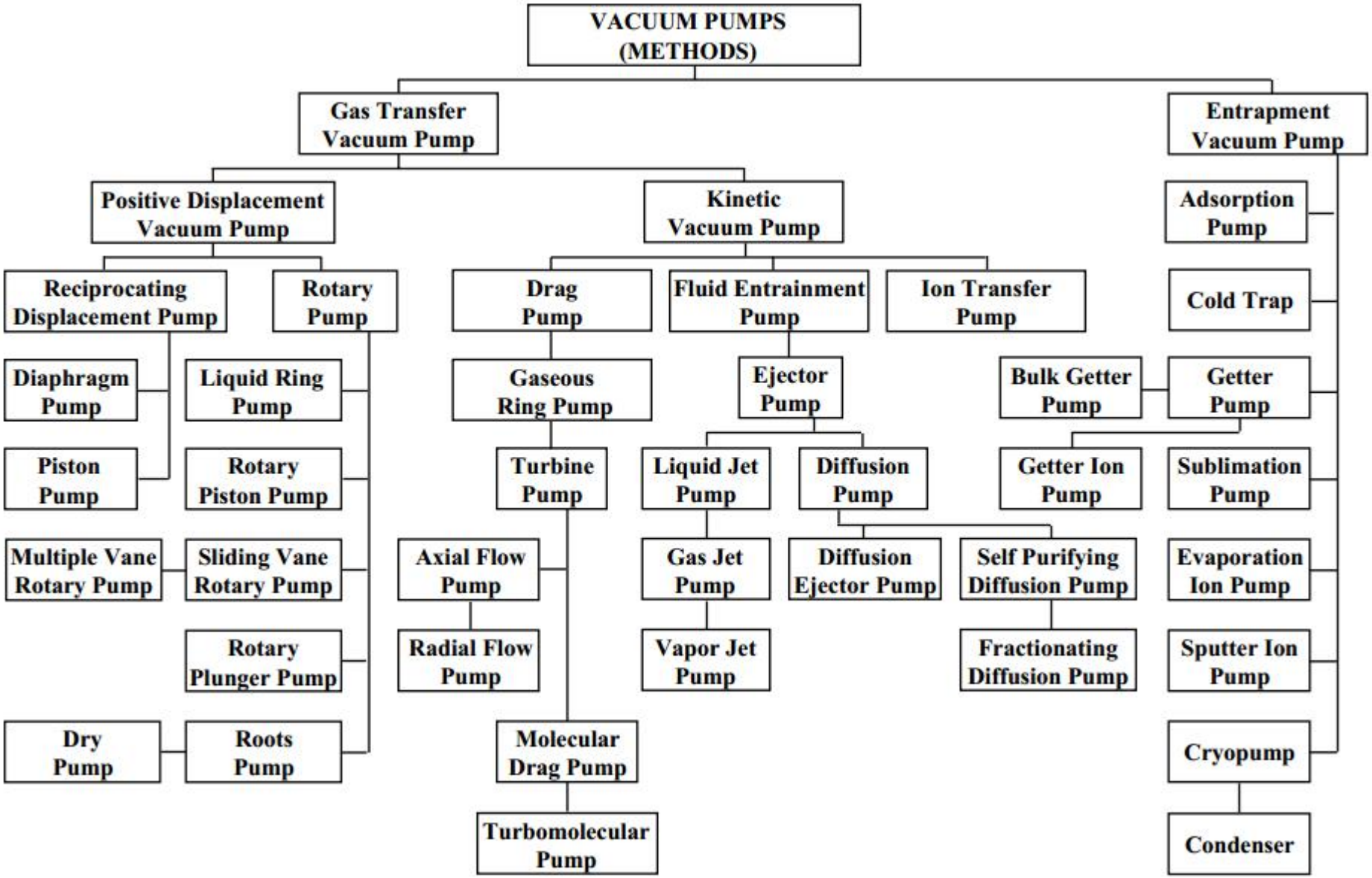
Gas-binding vacuum pumps have a limited gas absorption capacity and must be regenerated at certain process-dependent intervals

Gas-displacement pumps, which are also referred to as gas transfer pumps, are classified either as positive displacement pumps or kinetic vacuum pumps.

Positive displacement pumps displace gas from sealed areas to the atmosphere or to a downstream pump stage. Kinetic pumps displace gas by accelerating it in the pumping direction, either via a mechanical drive system or through an aligned vapor stream that is condensed at the end of the pumping section.

Gas-binding vacuum pumps either bind the gas to an especially active substrate through gettering or condense the gas at a suitable temperature.


Vacuum Pumps (Operating Principle)
Gas Transfer Vacuum Pumps
Gas-Binding Vacuum Pumps
Gas-Displacement Vacuum Pumps
Kinetic Vacuum Pumps
Oscillation Displacement Pumps
Double-Rotor Displacement Pumps
Single-Rotor Displacement Pumps
Mechanical Kinetic Pumps
Propellant Pumps
Ion Transfer Pump
Adsorption Pumps
Diaphragm Pump
Piston Pump
Scroll Pump
Roots Pump
Screw Pump
Claw-Type Piston Pump
Liquid Ring Pump
Rotary Vane Pump
Multi-Cell Pump
Rotary Piston Pump
External Vane Pump
Gas Ring (Side Channel) Pump
Turbopumps
Axial Pump
Radial Pump
Molecular Pump
Turbomolecular Pump
Propellant Jet Pump
Liquid Jet Pump
Liquid Jet Pump
Steam Jet Pump
Diffusion Pump
Diffusion Ejector Pump
Getter Pumps
Massive Getter Pump
Sublimation Vaporization Pump
Ion Getter Pump
Cryo Pump
Condenser


VACUUM PUMPS (METHODS)
Gas Transfer Vacuum Pump
Entrapment Vacuum Pump
Positive Displacement Vacuum Pump
Kinetic Vacuum Pump
Adsorption Pump
Reciprocating Displacement Pump
Rotary Pump
Drag Pump
Fluid Entrainment Pump
Ion Transfer Pump
Cold Trap
Diaphragm Pump
Liquid Ring Pump
Gaseous Ring Pump
Ejector Pump
Bulk Getter Pump
Getter Pump
Piston Pump
Rotary Piston Pump
Turbine Pump
Liquid Jet Pump
Diffusion Pump
Getter Ion Pump
Sublimation Pump
Multiple Vane Rotary Pump
Sliding Vane Rotary Pump
Axial Flow Pump
Gas Jet Pump
Diffusion Ejector Pump
Self Purifying Diffusion Pump
Evaporation Ion Pump
Rotary Plunger Pump
Radial Flow Pump
Vapor Jet Pump
Fractionating Diffusion Pump
Sputter Ion Pump
Dry Pump
Roots Pump
Molecular Drag Pump
Cryopump
Turbomolecular Pump
Condenser

Definition of volume flow rate, or pumping speed

Over a given inlet pressure range; i.e. they displace a constant volume flow.

$$S = \frac{dV}{dt}$$

Multiplying the volume flow rate by the inlet pressure yields the throughput of a pump.

Vacuum pump throughput

$$q_{pV} = S \cdot p = \frac{dV}{dt} \cdot p$$

Throughput is the gas flow transported by a vacuum pump.

Ultimate and Base Pressure

Ultimate pressure pe is the lowest pressure that is asymptotically approached by the pressure of a blank-flanged vacuum pump under defined basic conditions without gas inlet.

If a pump is operated at ultimate pressure, the usable pumping speed will be zero, as only its own backflow losses will be displaced. Ultimate pressure is a theoretical value.

Today, base pressure is specified instead of ultimate pressure. The conditions for achieving base pressure are specified in standard ISO 21360-1. As the base pressure must be attained within a specified period of time, it is usually higher than the ultimate pressure.

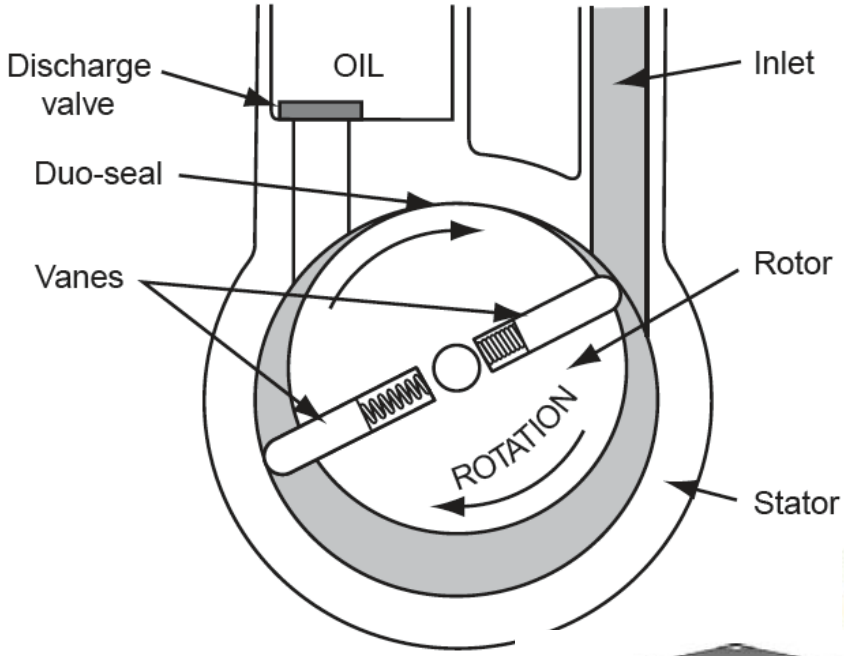
Compression Ratio:

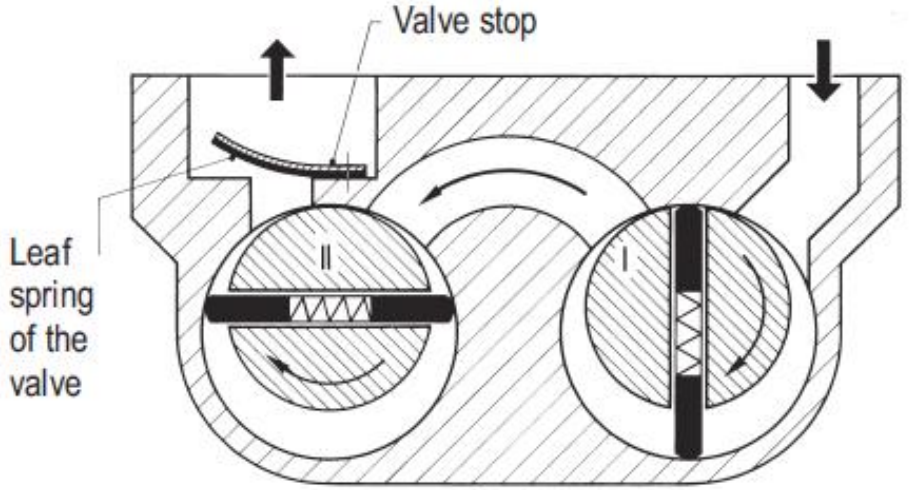
The maximum pressure ratio between discharge pressure *p2 and intake pressure p1 is* referred to as the compression ratio: $K_0 = p2/p1$

Pumping speed of pumping stages connected in series

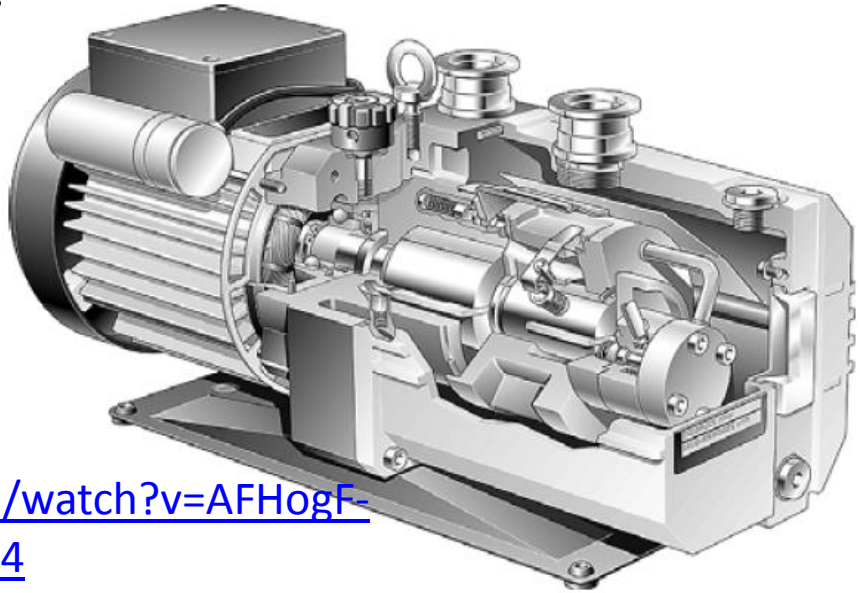
Let us consider a vacuum pump having a pumping speed *S0 and a compression ratio K0.*
The pump has backflow losses through gaps having conductivity *LR. Let inlet pressure be p1* and discharge pressure *p2. An additional pump having a pumping speed Sv is connected on* the outlet side.

Rotary vane vacuum pump





I High vacuum stage
II Second forevacuum stage



http://www.youtube.com/watch?v=AFHogF-9eGA&list=TLLtkRyN0XAk4

Design / Operating principle

A rotary vane vacuum pump is an oil-sealed rotary displacement pump. The pumping system consists of a housing (1), an eccentrically installed rotor (2), vanes that move radially under spring force (3) and the inlet and outlet (4).

The outlet valve is oil-sealed. The inlet valve is designed as a vacuum safety valve that is always open during operation.

The working chamber (5) is located inside the housing. Rotor and vanes divide the working chamber into two separate spaces having variable volumes.

As the rotor turns, gas flows into the enlarging suction chamber until it is sealed off by the second vane. The enclosed gas is compressed until the outlet valve opens against atmospheric pressure. In the case of gas ballast operation, a hole to the outside is opened, which empties into the sealed suction chamber on the front side.

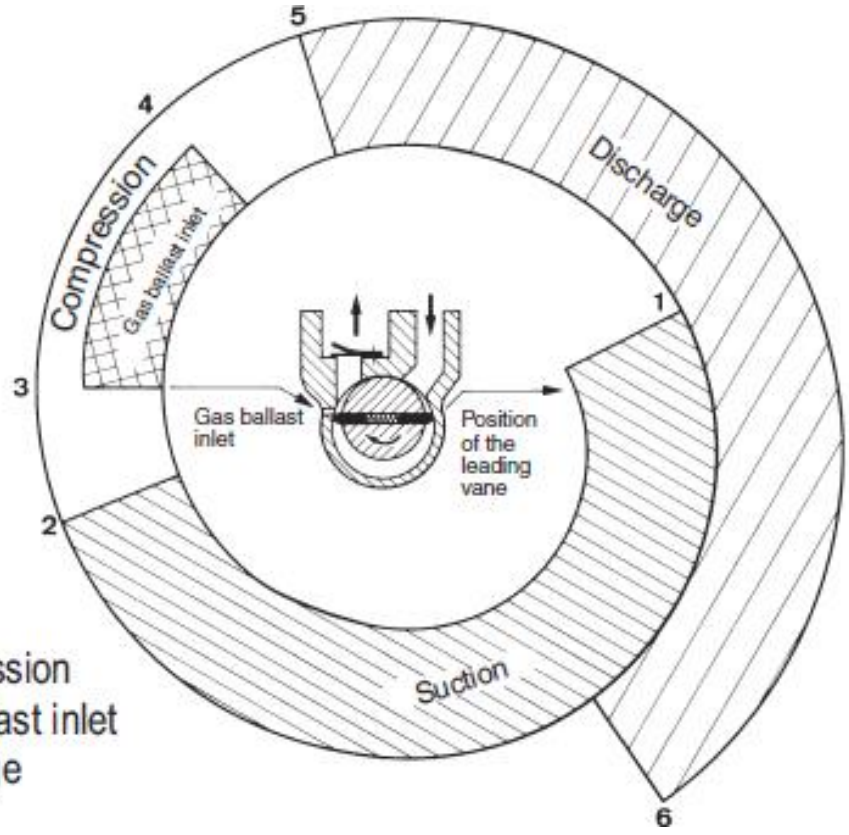
Operating fluid, oil

Pump oil, which is also called as operating fluid, has multiple tasks to perform in a rotary vane pump. It lubricates all moving parts, fills both the harmful space under the outlet valve as well as the narrow gap between inlet and outlet. It compresses the gap between the vanes and the working chamber and additionally ensures an optimal temperature balance through heat transfer.

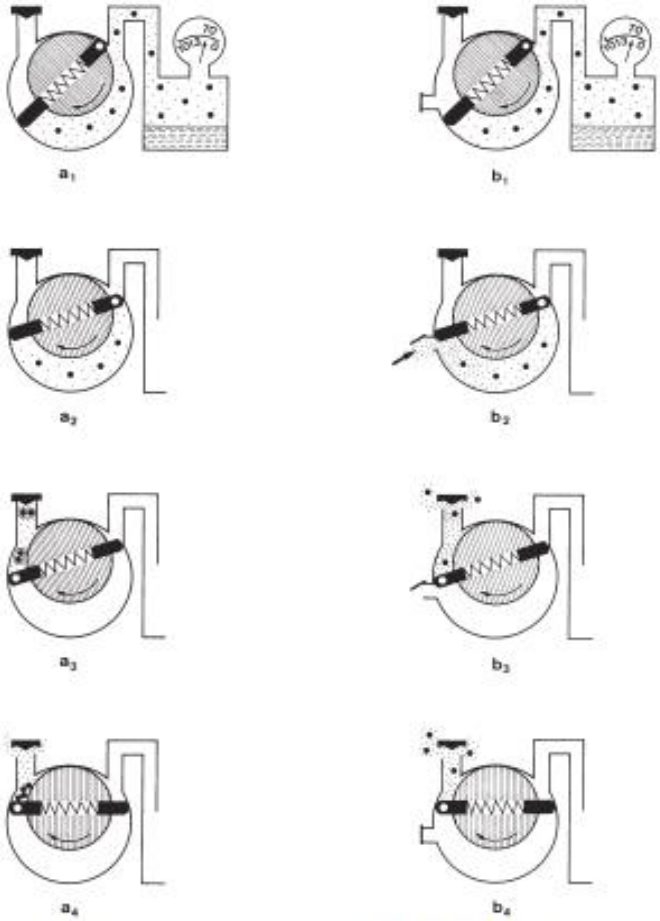
Multi-stage pumps

Rotary vane vacuum pumps are built in single- and two-stage versions. Two-stage pumps achieve lower ultimate pressures than single-stage pumps. Moreover, the effects of the gas ballast on the ultimate pressure are lower, as the ballast gas is only admitted in the second stage.

The gas ballast facility as used in the rotary vane pumps permits not only pumping of permanent gases but also even larger quantities of condensable gases. The gas ballast facility (see Fig. 2.13) prevents condensation of vapors in the pump chamber of the pump. When pumping vapors these may only be compressed up to their saturation vapor pressure at the temperature of the pump. If pumping water vapor, for example, at a pump temperature of 70 °C, the vapor may only be compressed to 312 mbar (saturation vapor pressure of water at 70 °C). When compressing further, the water vapor condenses without increasing the pressure. No overpressure is created in the pump and the exhaust valve is not opened. Instead the water vapor remains as water in the pump and emulsifies with the pump's oil. This very rapidly impairs the lubricating properties of the oil and the pump may even seize when it has taken up too much water. The gas ballast facility developed in 1935 by Wolfgang Gaede inhibits the occurrence of condensation of the vapor in the pump as follows. Before the actual compression process begins (see Fig. 2.13), a precisely defined quantity of air ("the gas ballast") is admitted into the pumping chamber of the pump. The quantity is such that the compression ratio of the pump is reduced to 10:1 max. Now vapors which have been taken in by the pump may be compressed together with the gas ballast, before reaching their condensation point and ejected from the pump. The partial pressure of the vapors which are taken in may however not exceed a certain value. It must be so low that in the case of a compression by a factor of 10, the vapors can not condense at the operating temperature of the pump. When pumping water vapor this critical value is termed the "water vapor tolerance".



1–2 Suction
2–5 Compression
3–4 Gas ballast inlet
5–6 Discharge



a) Without gas ballast

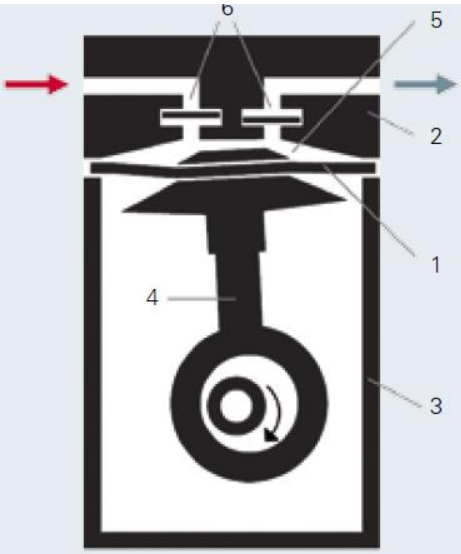
1) Pump is connected to the vessel, which is already almost empty of air (70 mbar) – it must thus transport mostly vapor particles
2) Pump chamber is separated from the vessel – compression begins
3) Content of pump chamber is already so far compressed that the vapor condenses to form droplets – overpressure is not yet reached
4) Residual air only now produces the required overpressure and opens the discharge valve, but the vapor has already condensed and the droplets are precipitated in the pump.

b) With gas ballast

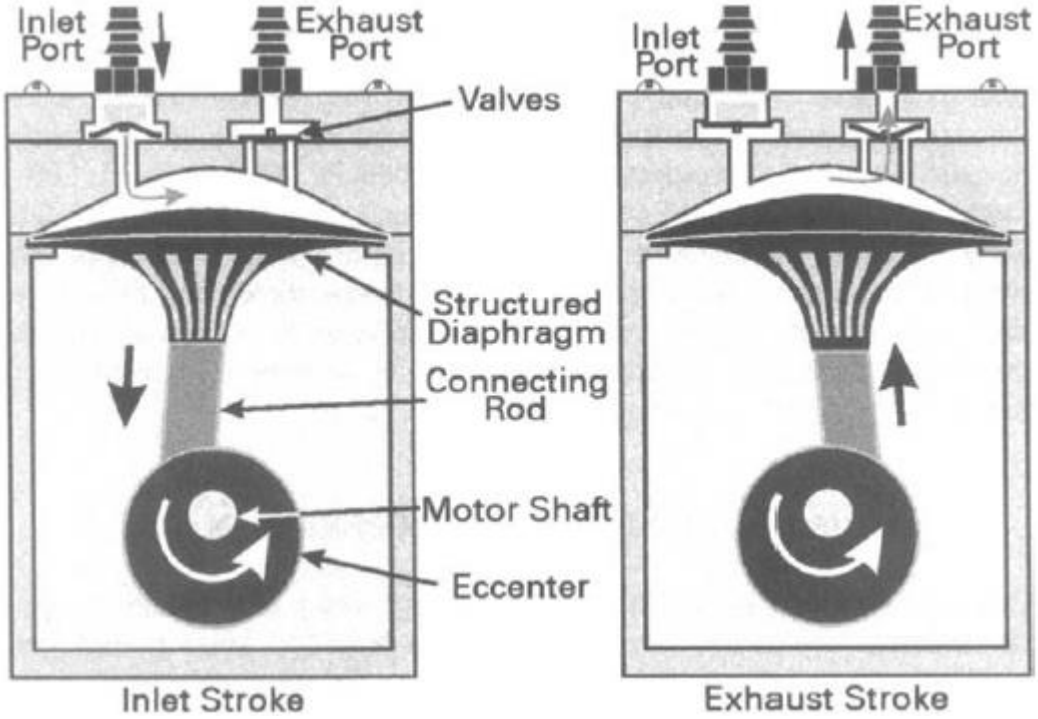
1) Pump is connected to the vessel, which is already almost empty of air (70 mbar) – it must thus transport mostly vapor particles
2) Pump chamber is separated from the vessel – now the gas ballast valve, through which the pump chamber is filled with additional air from outside, opens – this additional air is called gas ballast
3) Discharge valve is pressed open, and particles of vapor and gas are pushed out – the overpressure required for this to occur is reached very early because of the supplementary gas ballast air, as at the beginning the entire pumping process condensation cannot occur
4) The pump discharges further air and vapor

Diaphragm vacuum pumps

Diaphragm vacuum pumps are dry positive-displacement pumps. Their operating principle is explained in Figure 2.5. A crankshaft-driven connecting rod (4) moves the diaphragm (1) that is tensioned between head cover (2) and housing (3). The space between the head cover and the diaphragm forms the suction chamber (5). Diaphragm pumps require inlet valves and outlet valves (6) to achieve aligned gas displacement. Pressure-controlled shutter valves made of elastomer materials are used as valves. Because the suction chamber is hermetically sealed off from the drive by the diaphragm, the pump medium can neither be contaminated by oil nor can aggressive media corrode the mechanics. The harmful space between the outlet valve and the suction chamber results in only a limited compression ratio. This means that an ultimate pressure of only approximately 70 mbar can be attained with a single pump stage. Connecting multiple pumping stages in series can reduce ultimate pressure to 0.5 mbar. Lower pressures cannot be achieved, as in this case there is no longer sufficient force to open the inlet valve. The principle of the diaphragm pump is particularly well suited for low pumping speeds of up to approximately 10 m3 / h.



1) Diaphragm
2) Head cover
3) Housing
4) Connecting rod
5) Suction chamber
6) Valves



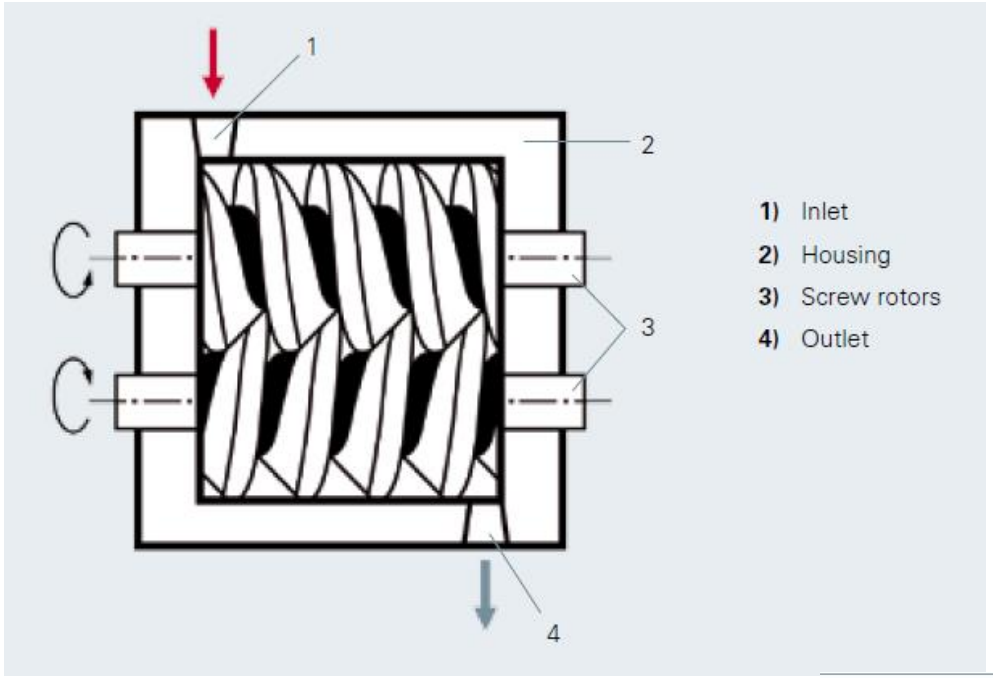
Diaphragm vacuum pumps are dry positive-displacement pumps.
A crankshaft-driven connecting rod (4) moves the diaphragm (1) that is tensioned between head cover (2) and housing (3). The space between the head cover and the diaphragm forms the suction chamber (5). Diaphragm pumps require inlet valves and outlet valves (6) to achieve aligned gas displacement.

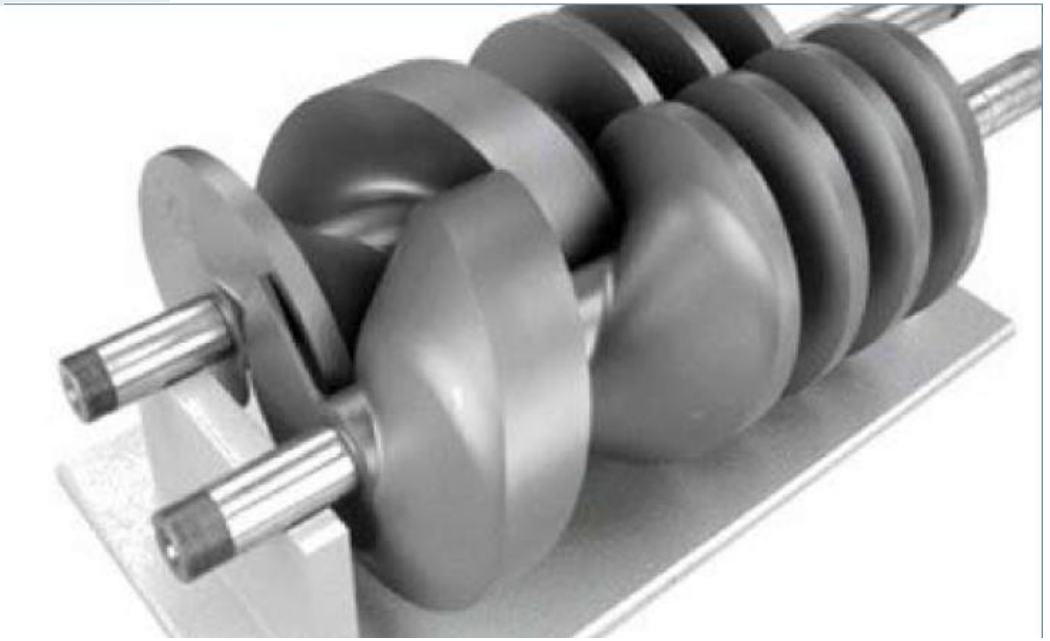
Pressure-controlled shutter valves made of elastomer materials are used as valves. Because the suction chamber is hermetically sealed off from the drive by the diaphragm, the pump medium can neither be contaminated by oil nor can aggressive media corrode the mechanics.

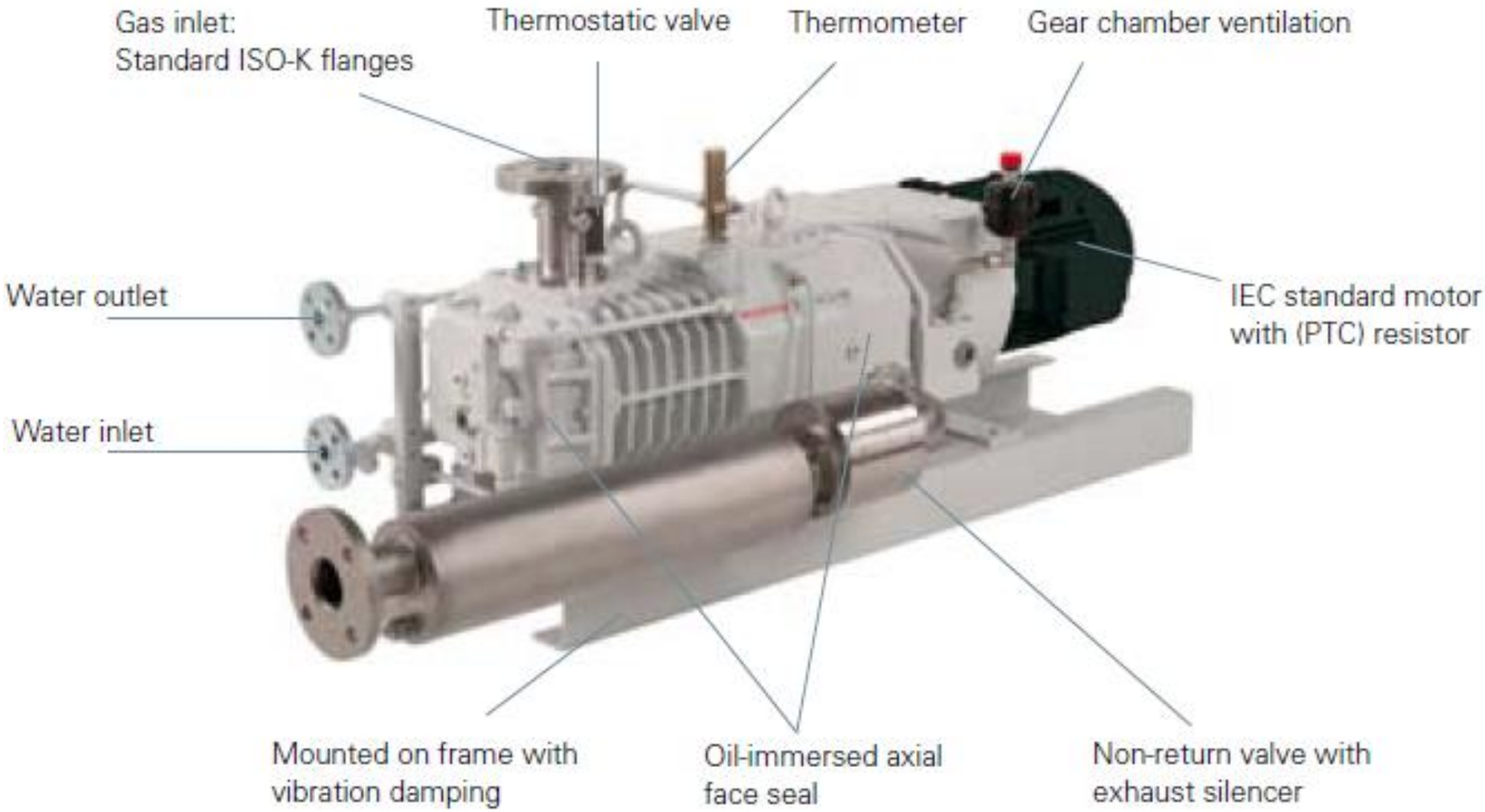
The harmful space between the outlet valve and the suction chamber results in only a limited compression ratio. This means that an ultimate pressure of only approximately 70 mbar can be attained with a single pump stage. Connecting multiple pumping stages in series can reduce ultimate pressure to 0.5 mbar.

Lower pressures cannot be achieved, as in this case there is no longer sufficient force to
open the inlet valve. The principle of the diaphragm pump is particularly well suited for
low pumping speeds of up to approximately 10 m3 / h.

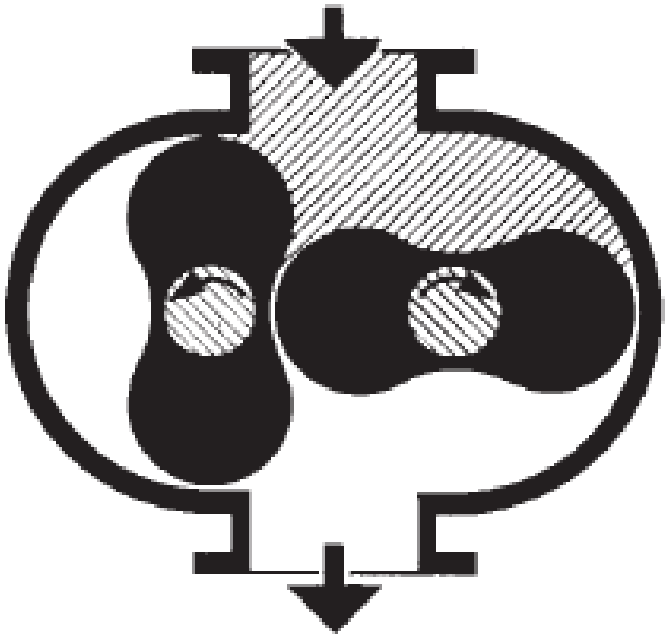
Screw vacuum pumps






Gas inlet:
Standard ISO-K flanges
Thermostatic valve
Thermometer
Gear chamber ventilation
Water outlet
IEC standard motor
with (PTC) resistor
Water inlet
Mounted on frame with
vibration damping
Oil-immersed axial
face seal
Non-return valve with
exhaust silencer

Roots vacuum pumps



1 Intake flange
2 Rotors
3 Chamber
4 Exhaust flange
5 Casing

The Roots pumps used in the low and medium vacuum ranges with high volumetric flow.

It cannot exhaust to atmospheric pressure and is operated in series with a backing pump, often of the rotary vane variety, or 'dry' pump, connected to the exhaust. The Roots pump has two lobed rotors interlocked and synchronised by external gearing which rotate in opposite directions, moving past each other and the chamber with small clearance (0.3 mm). The gas is displaced from the inlet to the outlet.

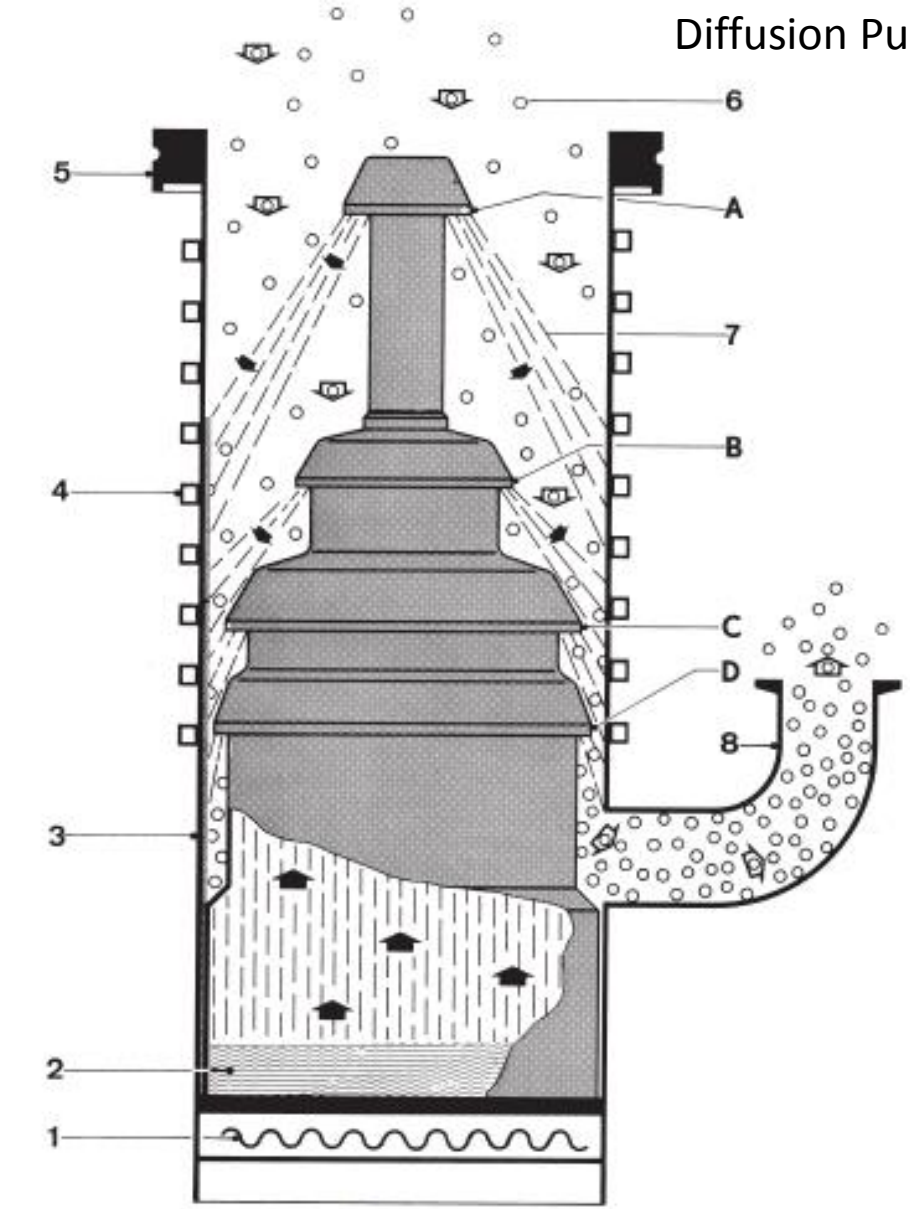
Scroll pump

http://upload.wikimedia.org/wikipedia/commons/3/31/Two_moving_spirals_scroll_pump.gif

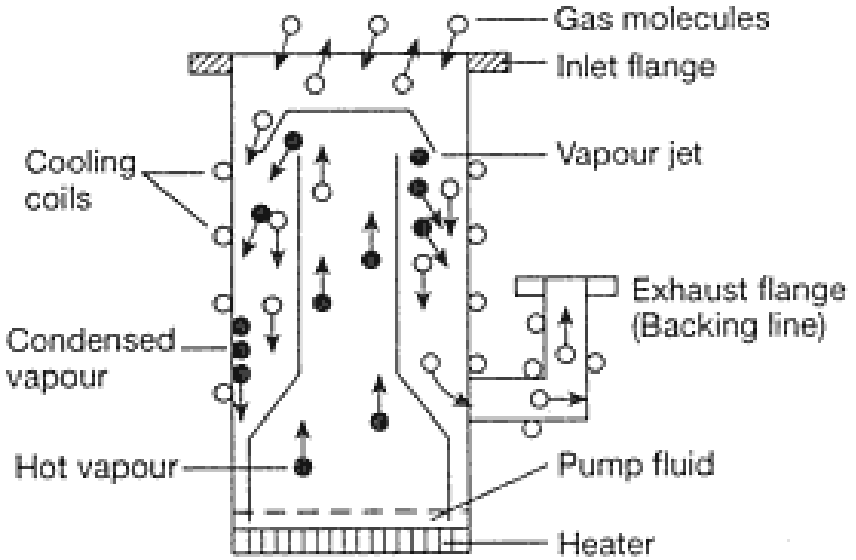


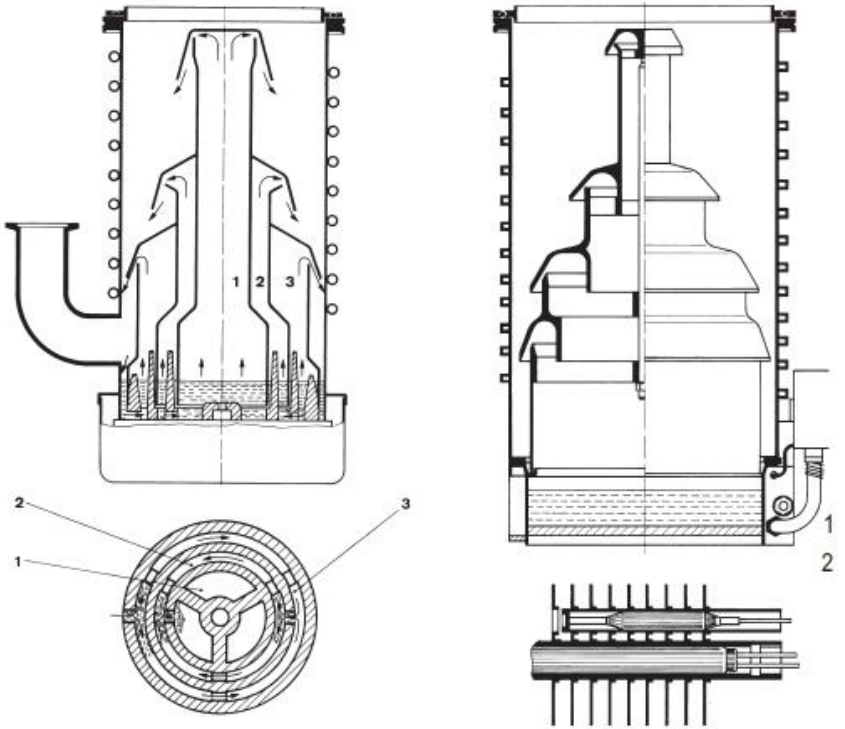
The scroll pump consists of an orbiting (not rotating) scroll meshing with a fixed scroll. This movement causes pockets of gas to be compressed and passed round to the centre of the pump and discharged to the atmosphere. It has no lubricating or sealing fluids, so it is a dry clean pump. It has an ultimate pressure of 10-2 mbar and can be used as a backing pump for turbo- or cryo-pumps. It has been used to provide a stage in the pump-out of a sputter ion-pump.

Diffusion Pump

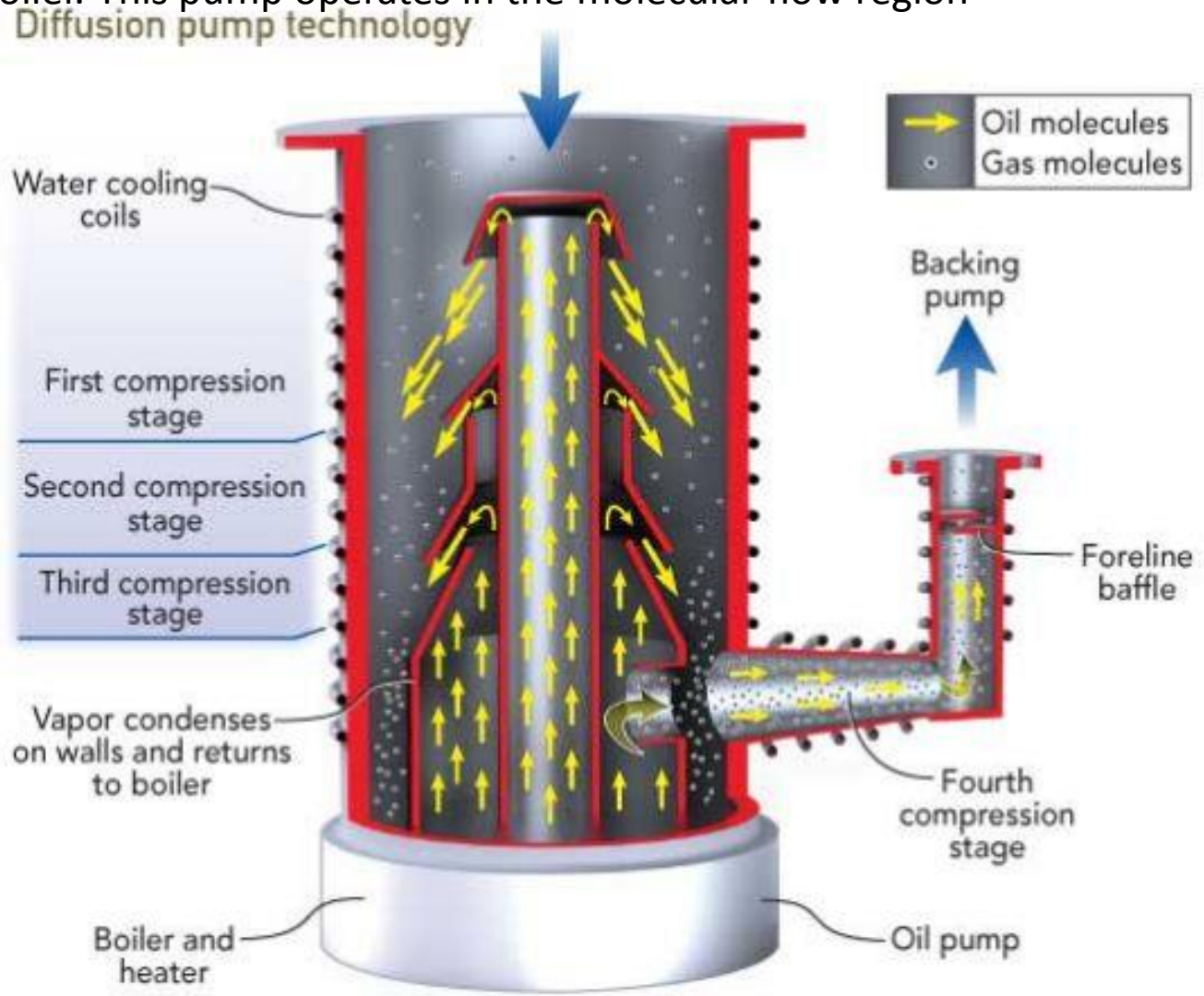


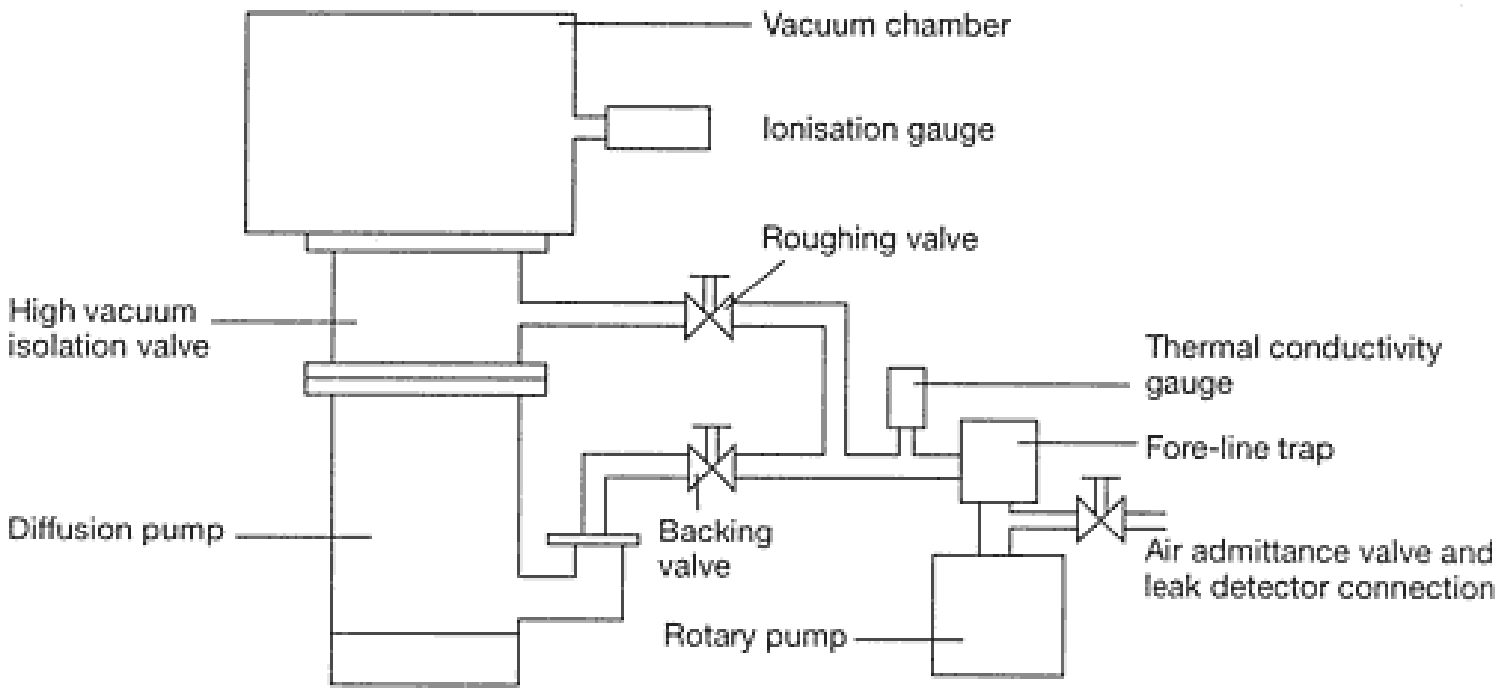
1 Heater
2 Boiler
3 Pump body
4 Cooling coil
5 High vacuum flange
6 Gas molecules
7 Vapor jet
8 Backing vacuum connection
A, B, C, D } Nozzles





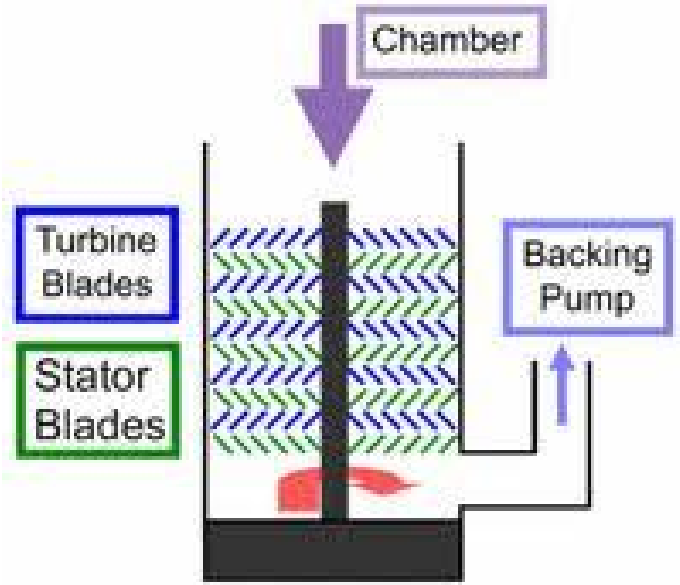
In this pump a heater vaporises the working fluid which rises inside the vapour chimney and is deflected downwards by the jet assembly in an annular jet at supersonic speed. This high speed jet of fluid molecules imparts a momentum to the random moving incoming gas molecules giving them a direction primarily towards the region of the pump outlet where they are removed by a mechanical pump. The vapour jet condenses on the cooled pump walls and returns to the boiler. This pump operates in the molecular flow region

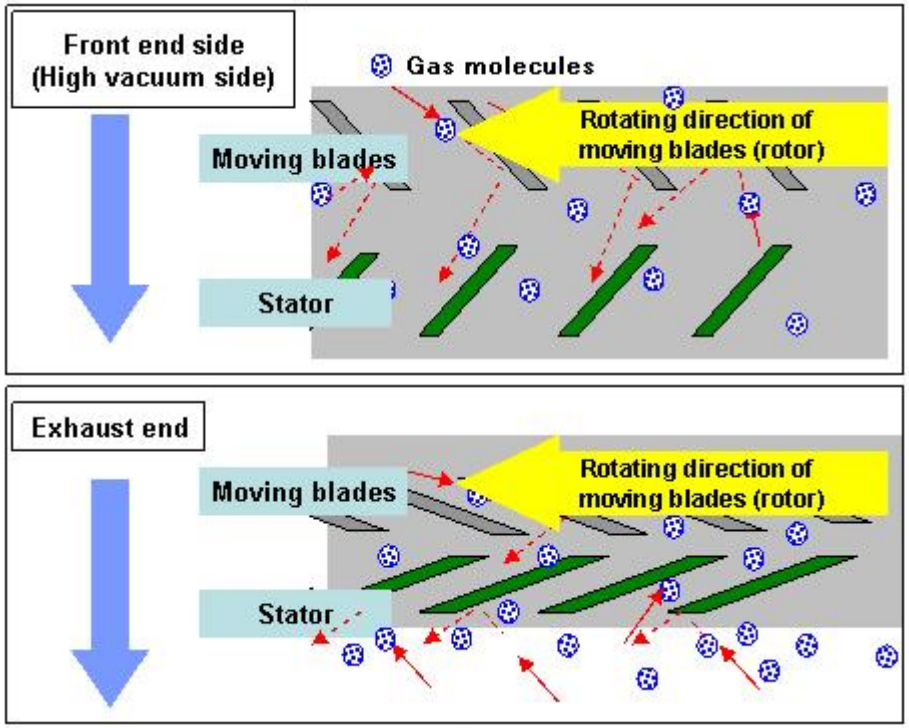



Vacuum chamber
Ionisation gauge
Roughing valve
High vacuum
isolation valve
Thermal conductivity
gauge
Fore-line trap
Diffusion pump
Backing
valve
Air admittance valve and
leak detector connection
Rotary pump

Turbo Molecular Pump

The pumping effect of an arrangement consisting of rotor and stator blades is based upon the transfer of impulses from the rapidly rotating blades to the gas molecules being pumped. Molecules that collide with the blades are adsorbed there and leave the blades again after a certain length of time. In this process, blade speed is added to the thermal molecular speed. To ensure that the speed component that is transferred by the blades is not lost due to collisions with other molecules, molecular flow must prevail in the pump, i.e. the mean free path length must be greater than the blade spacing.




Front end side
(High vacuum side)
Gas molecules
Rotating direction of
moving blades (rotor)
Moving blades
Stator
Exhaust end
Moving blades
Rotating direction of
moving blades (rotor)
Stator

